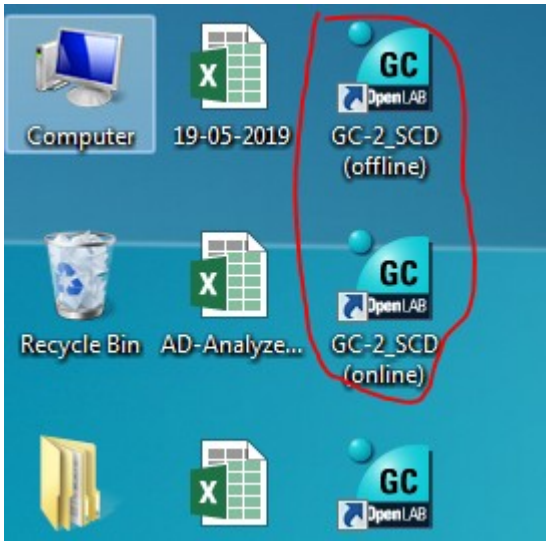
Ilya Kutuzov last updates: 8/11/2020; 14/06/2022; 5/11/2025

Before you start- The GC-FPD you want to use is custom-made, contains many one-of-a-kind sulfinert-coated parts, uses several types of columns and contains in-house upgrades. If you break something it could be VERY difficult to fix or replace. If you are not sure of what you are doing, ask Alon and seek more information at the "Agilent 7890B manual" (google it).

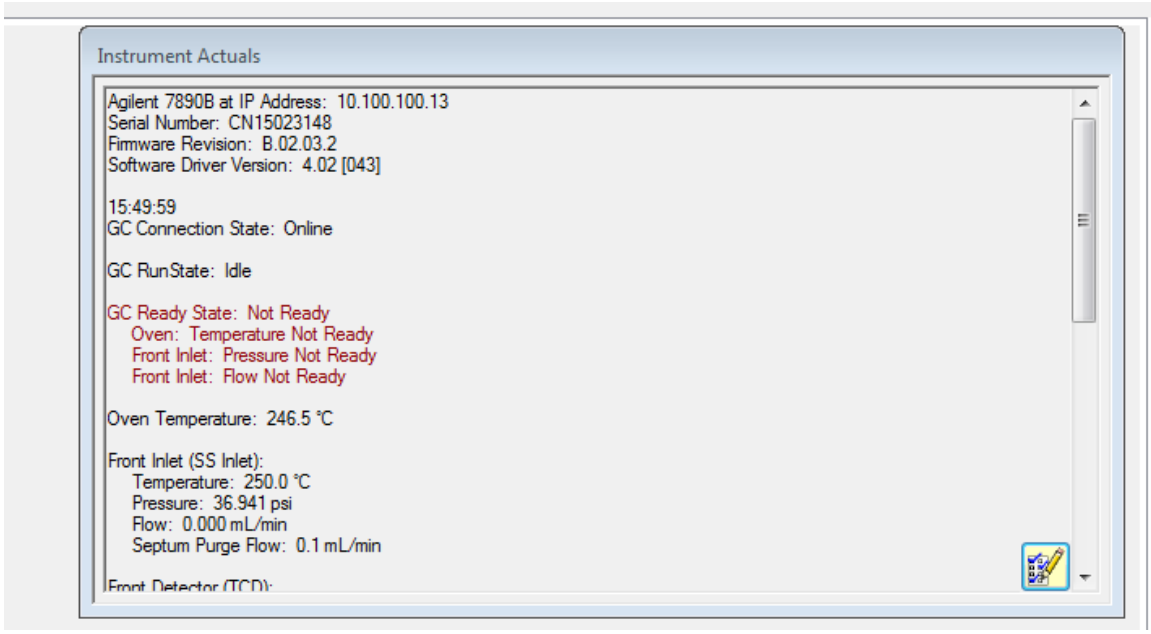
Ilya Kutuzov last updates: 8/11/2020; 14/06/2022; 5/11/2025

How to perform analysis on the GC-DFPD:

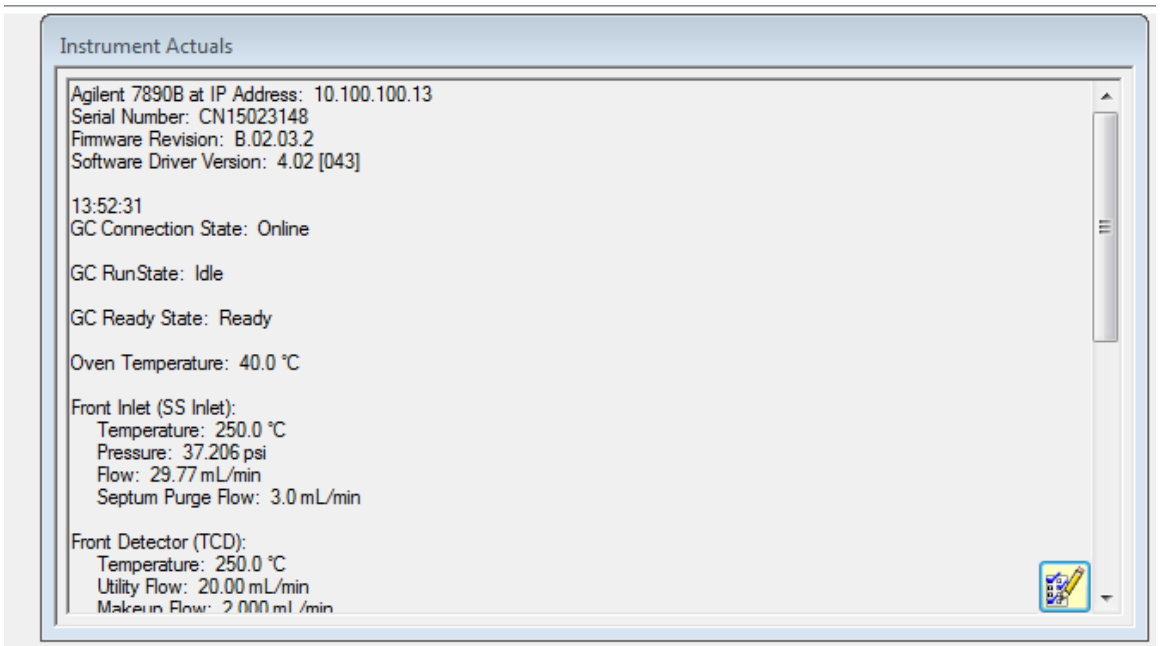
1. Before starting: make sure you have enough gases to work. You will need He (99.995%), dry air, N_2 (99.99%) and H_2 (from the hydrogen generator under the table). The hydrogen generator needs DDW to operate, make sure there is enough DDW in it (canister of DDW is behind you next to the left fridge). Open the cylinders now, make sure you open the small valve on the regulator exit.
2. Start the GC-2_SCD (2025 update: its "7890DFPD" now) **online program and then offline program**. Online is used for operation of the GC while offline is used for data analysis. Two popup windows (regarding method selection and data folders) should appear, press esc (on the keyboard) both times.



3. Check that the connection between the GC and the computer is established by using the "Instrument Actuals" panel. If everything is fine you should see all the GC parameters in this window similar to this:

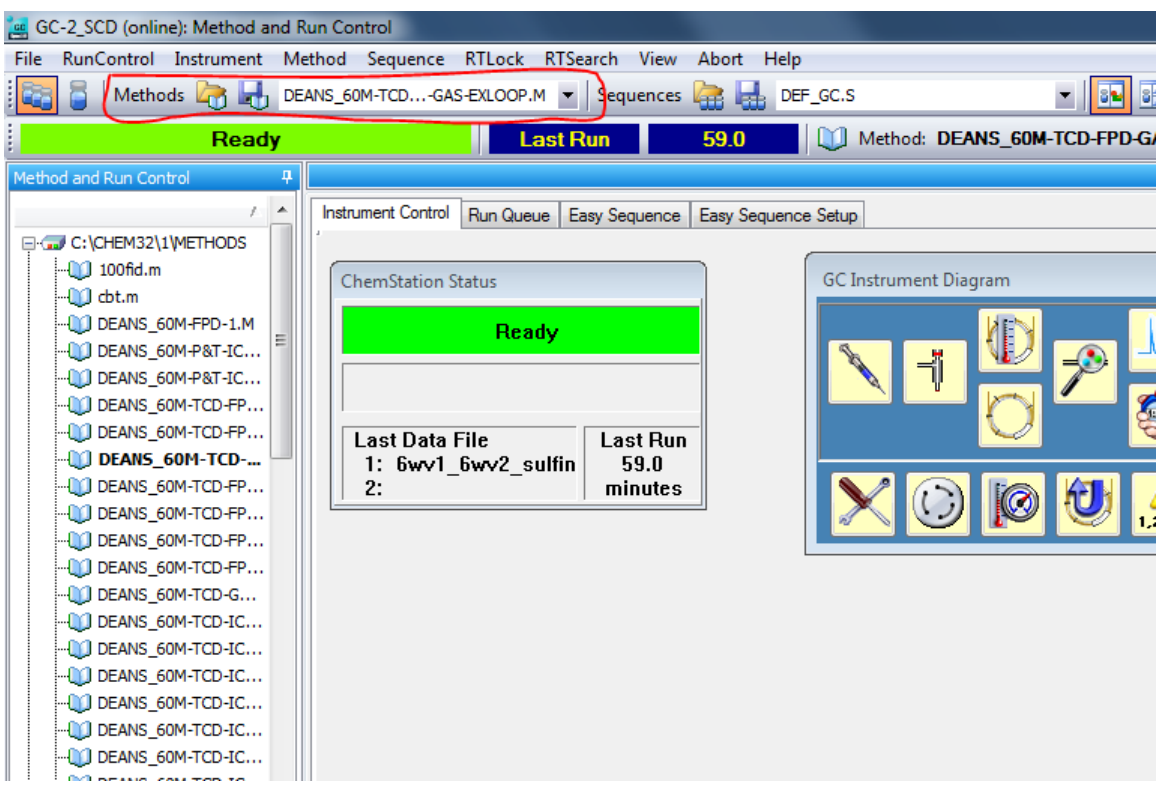


All of the red lines will disappear when the GC is ready:



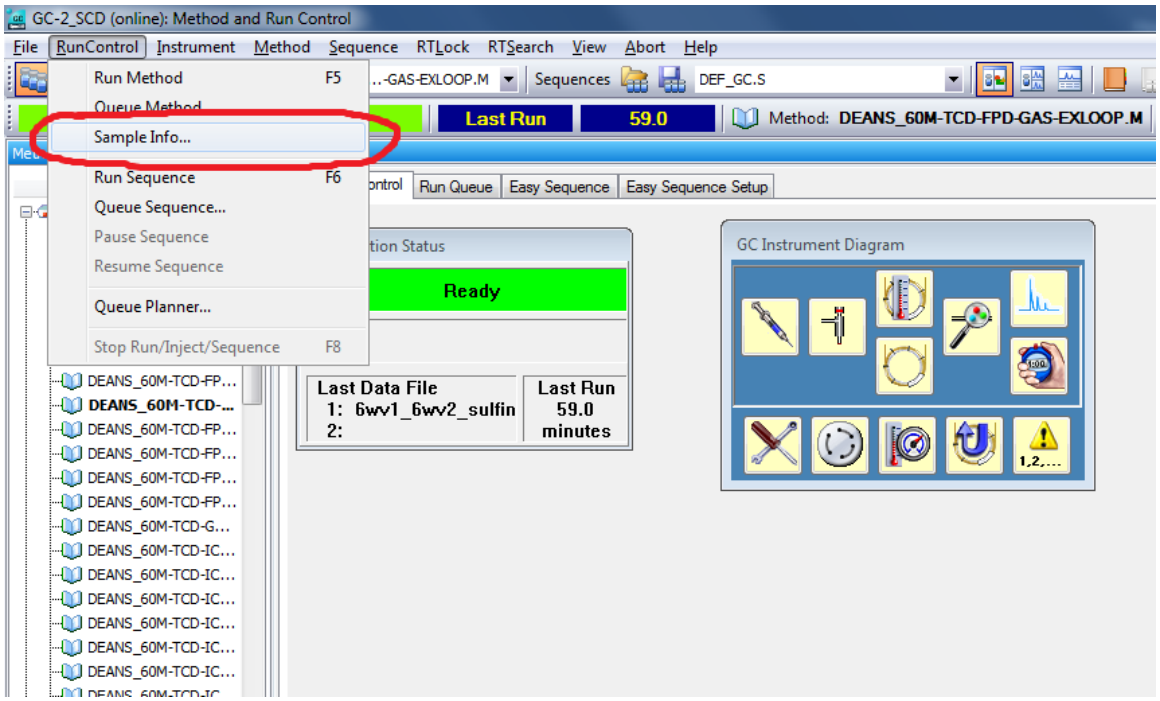
Otherwise, the window will be empty as there is no connection. To solve this problem, make sure the LAN cable between the GC and the computer is ok and restart the program until it works. If nothing works turn off the GC, unplug the black power line for a minute and then connect everything back together.

4. The last method used will appear in the method line, usually that would be "DEF_GC.M" which is used to cool down the GC prior to shut down. Chose the method you need from the list. The methods I created are divided to two groups: methods that use the inlet injection system (aka "direct-injection") (TCD_FPD_Gas or TCD_ICP_gas) and methods that use the exloop system ("TCD-FPD-Exloop"). The latter are used for the Tenax project, all injections with external sampling loops and all injections with liquid nitrogen trapping. If for some reason you want to inject with a syringe bia the inlet septa make sure you use the "TCD_FPD_Inj" method. In 2024/5 I switched to the Gaspro column for the OCS project. All new methods are labeled with "gaspro" in their method name.



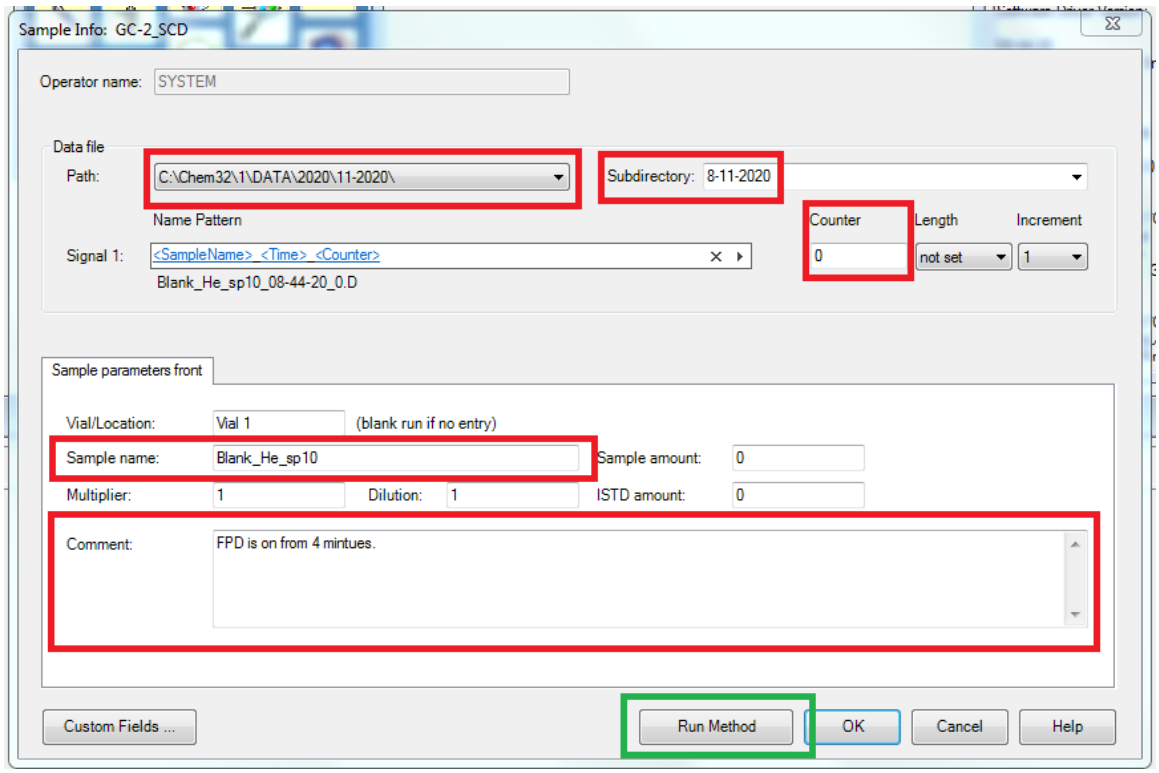
Allow the GC to heat up until the READY panel is green and the “Not-Ready” LED on the GC panel is off. Perform leak check inside the GC oven- this must be done each time you turn the GC on. After ignition of the FPD you must wait around 15 minutes for the detector to stabilize.

5. To perform an analysis chose “RunControl” and then “Sample info..”

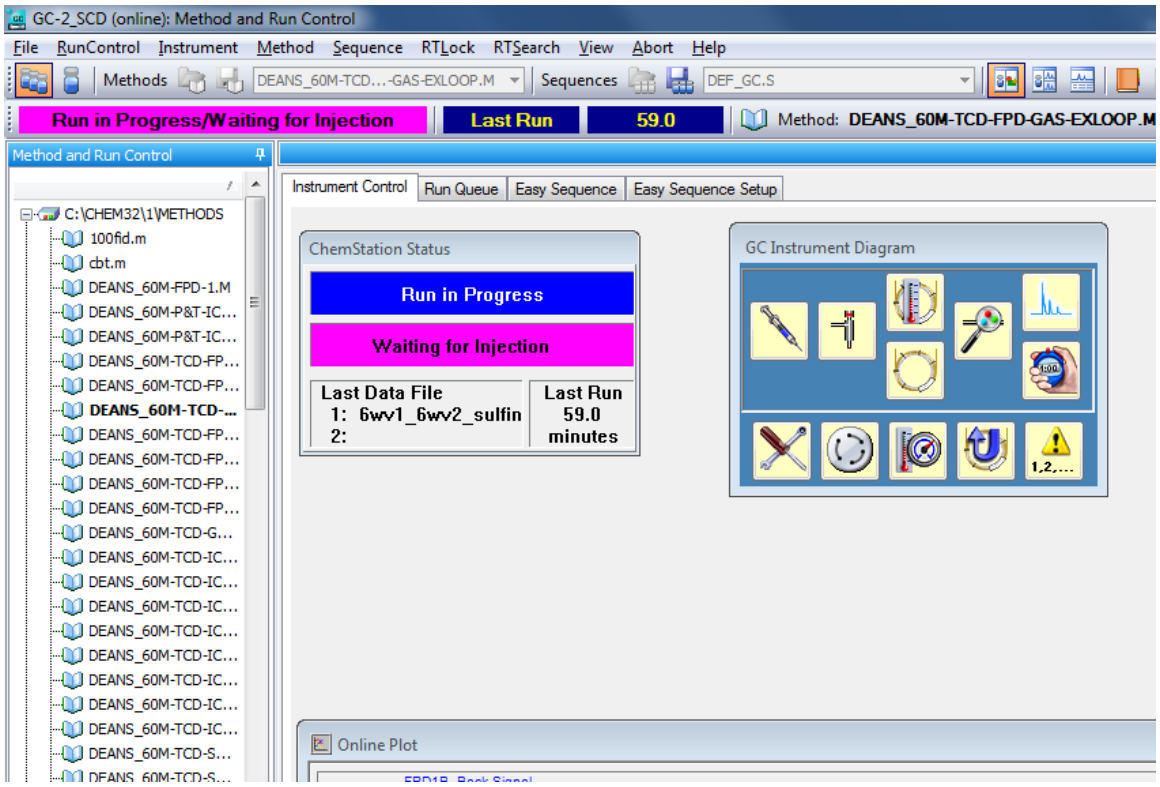


6. The “Path” is where the data file will be saved, make sure it’s the right month and year. If this is the first time using the GC this month you will need to prepare a new folder prior (more on that below). If it’s a new day in an already exciting month it can be done directly through the GC by filling the date in the “Subdirectory”, please use the dd-mm-yyyy date system. “Signal 1” is the overall final name of the sample and it consists of sample name, the time of injection and the counter. The sample name is filled below, try to provide as much info as possible, for example: use “std” for standard and “blank” for blank, “Re” for reactor, state the split etc... The “<counter>” part in the name refers to the sample

number in the given day so it must start 0 at the first sample of the day, the software will then automatically add 1 to each sample (no need to change that). The bottom box of comments is meant for short description of the sample, state only what's not obvious from the sample name. Press "Run Method" to start the analysis, **do not press** "OK" as it will only save the current sample settings but won't start the analysis.

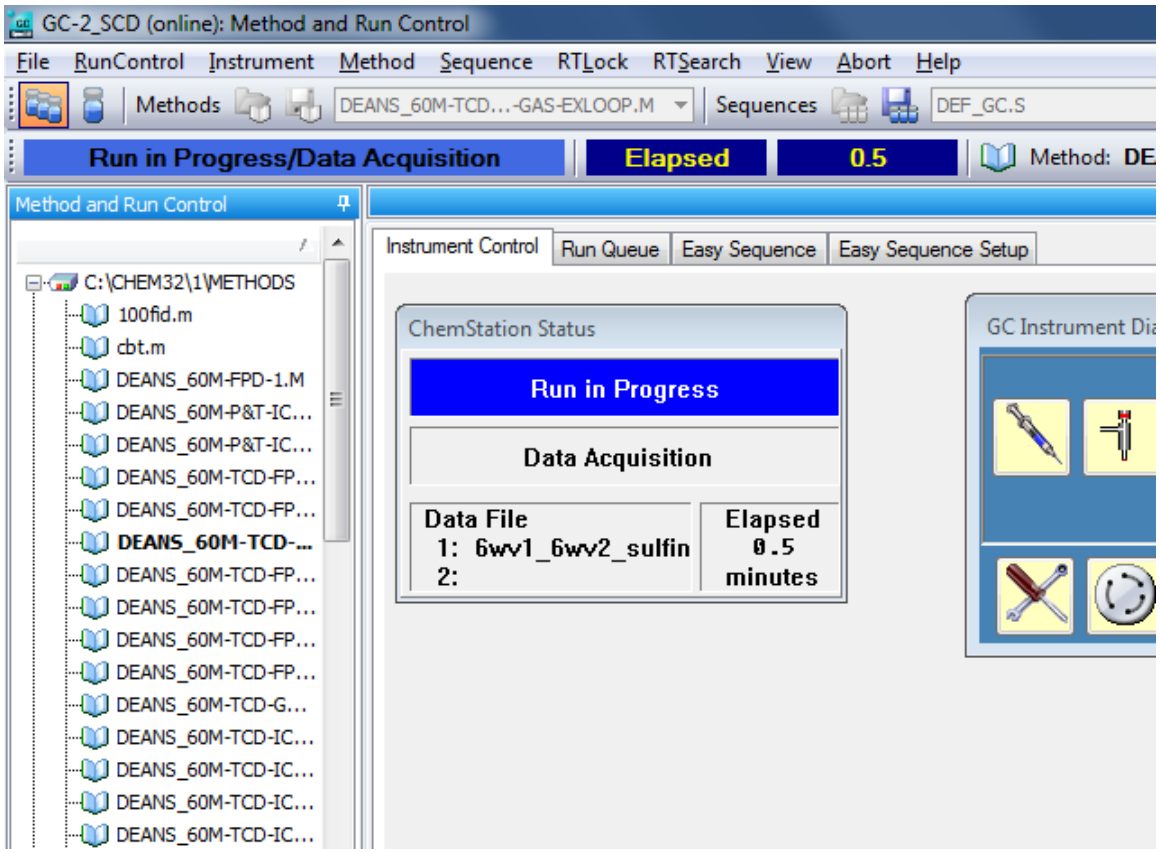


7. After pressing "Rum Method" the "ChemStation status" will look like this:



Now the analysis will only start when you press the “Start” button on the GC panel. The typical injection scheme for gases (through the gas inlet) is 15 seconds sample wash through the gas inlet, press “Start” and wash for another 45 seconds so that total wash time is 1 minute. During this 1 minutes you should hear the gas sampling valve (“Valve 1”) activate. Typical scheme for fluid injection is to press “Start” and immediately inject 1µL of sample by syringe to the inlet. The typical scheme for injection via external loop is to press “Start” and simultaneously switch the 6WV to position B and immerse the frozen sample loop into hot water for 15 seconds. After 15 seconds the 6WV is switched back to A.

Upon pressing “Start” the window will look like this:



The analysis is on, good luck!

How to turn off the DFPD:

1. Make sure the last analysis is done: the GC has cooled down back to its starting temperature (typically 40°C).
2. Select the method "DEF_GC", this method is used for controlled cool down of the GC system. When selected you will hear a click and the gas flow to the GC will stop.
3. Press the "Front inlet" button on the GC panel and choose "Pressure", turn it on so that He flow through the system. This is done to allow cooling of the column and sulfinert parts of the GC in an inert atmosphere. Not doing it will result in damage.
4. Close the gas sources to the GC: N_2, dry air, H_2 and other gas cylinders for exloop etc..
5. Turn off all heating tapes (such as those used for the 6WV's). Unplug the tapes from the socket.
6. When the temperature of the front inlet is <100°C turn off the pressure of the front inlet and close the He cylinder.
7. If you select the method "DEF_GC" and nothing happens (this is the situation in 2024/5) you will have to cool the GC through its instrument panel by manually switching off the following parameters (<u>in this order</u>):
    - Oven: Temperature (off)
    - Front Det: Filament (off), Makeup(He), Reference flow, temperature
    - Valve #: Valve box

- Back Det: Flame, PMT voltage, Makeup (N2), Air flow, H2 flow, Temperature
- Col 2: Pressure (only when front inlet temperature is <100°C)
- Front Inlet: Temperature, Pressure (only when front inlet temperature is <100°C)

Troubleshooting:

- After selecting the method of choice the GC screen displayed an error message saying back detector failed to ignite. This situation happens when the FPD flame does not turn on (2025 update- this is considered normal now and happens all the time). To solve this issue, press the "Back det" button on the GC panel, scroll down to the "Flame" and press the "Off/ No" button. Now scroll back up to the "H2 flow" and "Air flow" and turn both of them on by pressing "On/Yes" button. Also make sure the temperature of the detector is 200°C.

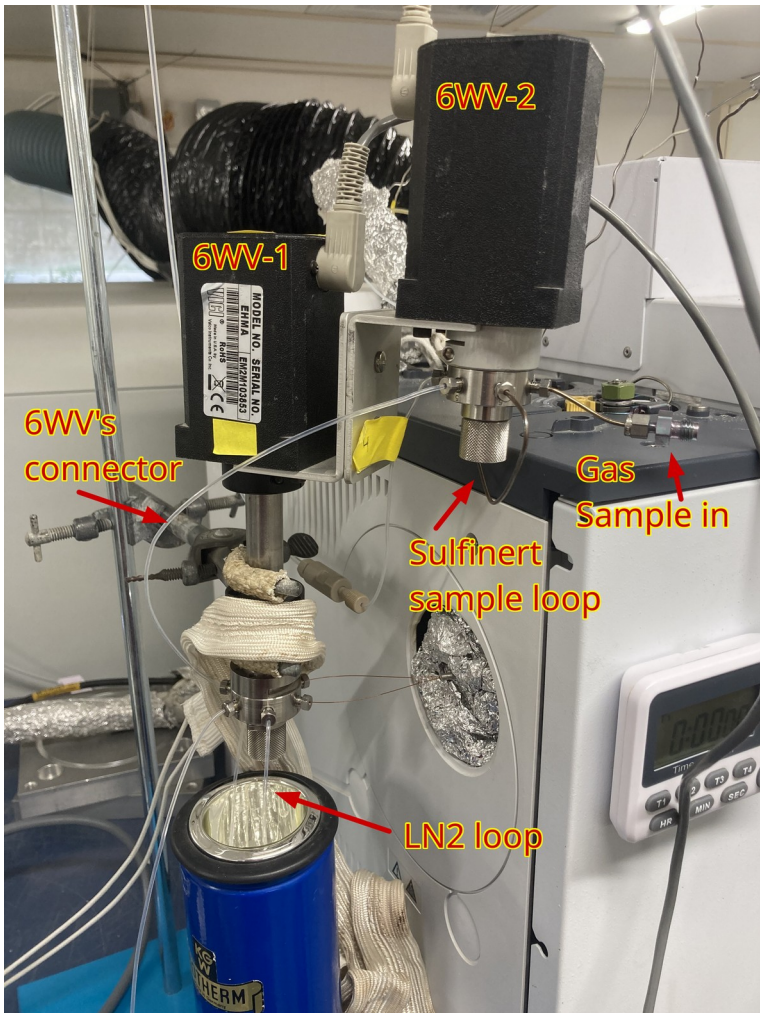
  Now you need to wait ~30 minutes to let the detector heat up and you can try turning it on by scrolling down to "Flame" and pressing "On/Yes" button. If fails, repeat.

  If the detector does not turn on after 5 attempts, it could be related to column breaking in the detector or other physical issues- seek advice from someone who knows how to replace the column section in the detector.

Using the external loop system ("exloop")

What is the exloop system:

The exloop composed on two 6 way valves (VICI, model EHMA). The first valve is used to collect a precise amount of gas within a sampling loop, the second valve is used to introduce the collected gas into the GC column.



The sampling 6 way valve ("6WV-2") has the following connections:

- Port 1: Sample in. Sulfinert line (1/16) with vici nut on one side and swagelok 1/16 nut on its other side. The swagelok nut is typically connected to a 1/16 to 1/8 reducer from sulfinert. The 1/8 is standard in the lab and easy to connect to gas sampling cylinders.

- Port 2: Vent. This goes to a flow meter to visually present the rate of sample flowing through 6WV-2. This is typically a metal 1/16 with vici nut on one side and swagelok 1/16 on the other side. The material of the line is not important because it goes to waster anyway.

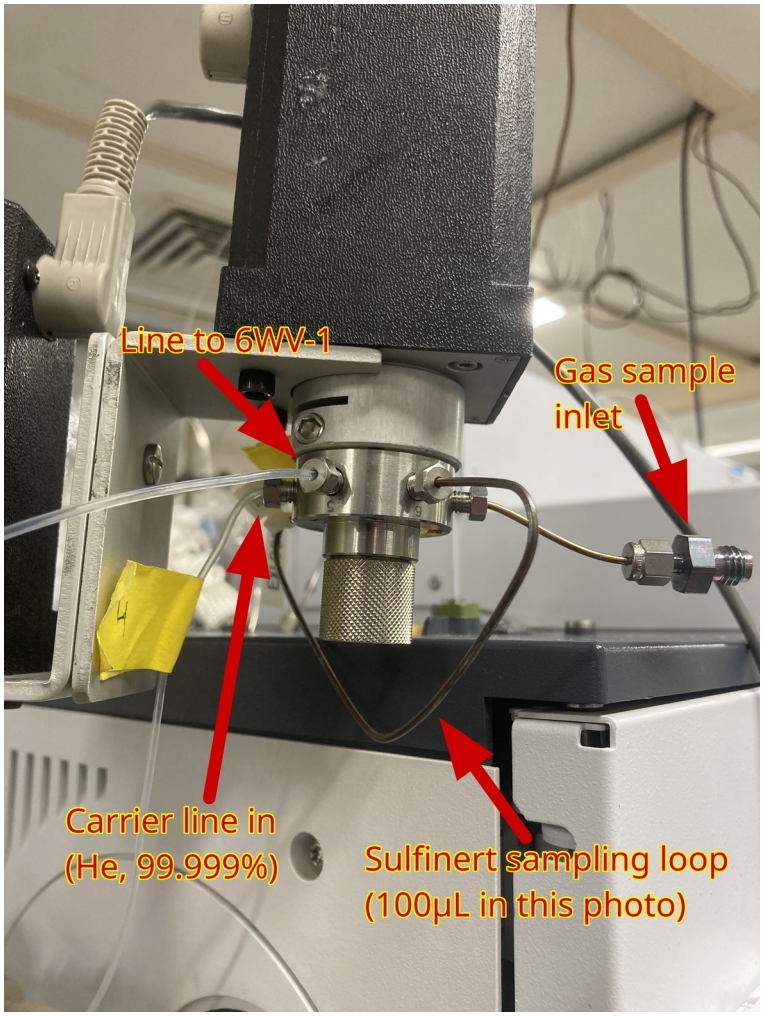
- Port 3: Sampling loop connection 1. The sampling loop must be sulfinert with known volume.

- Port 4: Carrier in. Typically He 99.999% at 10-20 ml/min. To adjust the rate a flow meter should be used.

- Port 5: Sample out. This line goes to the second 6WV. It should be the shortest possible line (minimal volume principle) to reach the other valve. I typically use

1/16 PTFE although sulfinert can be used as well. Both ends of the line have vici nut.

- Port 6: Sampling loop connection 2.



The introduction/injection 6 way valve ("6WV-1") has the following connections:

- Port 1: Sample in. This is where the the line from 6WV-2 is connecting to.
- Port 2: Vent. If everything works right there is nothing to be vented from this end so I just use a random 1/16 PTFE line. To save costs only one end of the line is fitted with a vici nut.

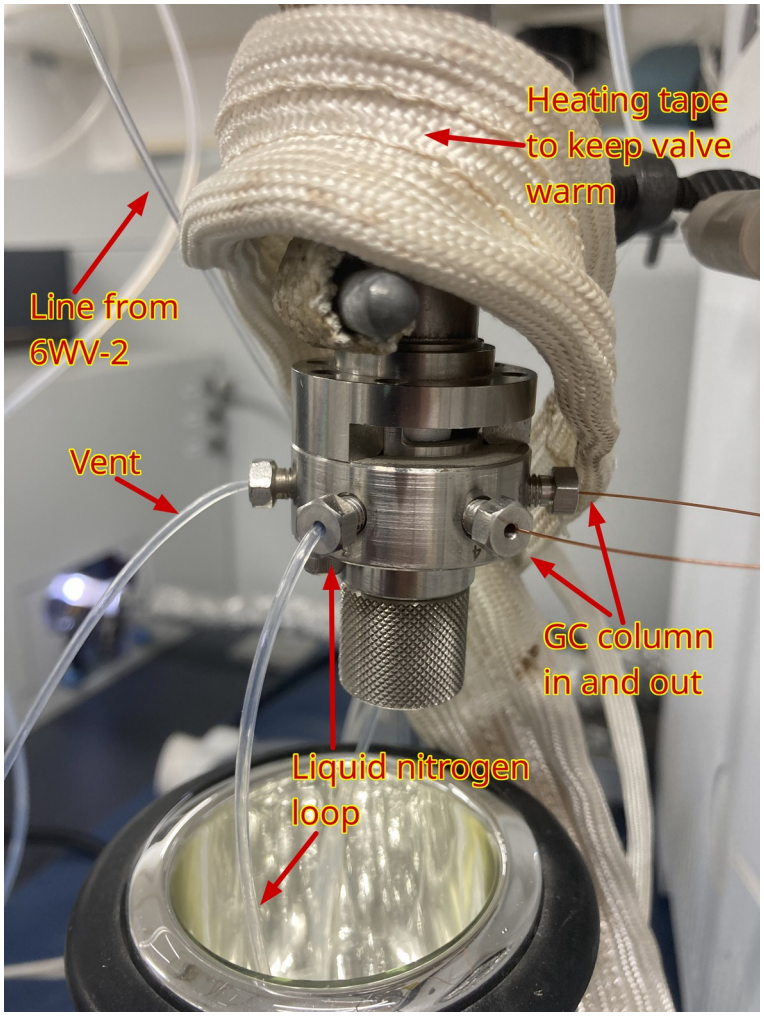
- Port 3: Liquid nitrogen loop connection 1. I typically prepare a loop from 1/16 PTFE with vici nuts on both sides. The length should be sufficient submerge most of the loop in liquid nitrogen within a dewar.

- Port 4: Column in. The GC column from the front inlet enters the exloop here. This part does not separate anything, it only carries the He from the GC to the exloop. Make sure you use the right vici ferrule between the valve and the vici

valve. Dont over-tighten the nut and make sure for leaks- this is the weak spot of the system.

- Port 5: Column out. The GC column itself. This is the part where the sample from the exloop enters the actual GC column. Same caution is for port 4.

- Port 6: Liquid nitrogen loop connection 2.



Standard injection:

1. Make sure there are no leaks in the exloop system by using both leak detector (check all connection from He cylinder to the GC column) and flow meter (make

sure flow at the vent is identical to the He flow to the system- flow in the system should be 10-20 ml/min).

2. Fill up the liquid nitrogen Dewar and place it underneath 6WV-1 so that the Teflon loop is in the Dewar.
3. Make sure 6WV-2 is in A position and 6WV-1 is in the A position. The A position is used to capture std/sample/whatever.
4. Slowly open the std by moving the cylinder valve anti-clockwise. Make sure that the marbles are around the 2-3 mark.
5. When you have stabilized the flow of std to the system start your timer for 20 sec. 2025 update- this 20 seconds mark is because this allows us the wash the whole system a huge amount of times. In practice, shorter was times down to 5 seconds were tested and seem to work.
6. After 20 seconds firstly change the 6WV-2 to B this will free the std to flow towards 6WV-1. Then close the std flow moving the dial clockwise.
7. Wait for 3 minutes to allow trapping of the sulfinert loop contents on the Teflon loop in the LN2.
8. After 3 minutes simultaneously press 'START' on the FPD panel and change 6WV-1 to the B position releasing the std toward the FPD.
9. Remove the liquid nitrogen and replace it with a mug filled with hot water. Hold for 15 seconds.
10. After 15 seconds remove the mug and switch the 6WV-1 back to A.

Typical analysis and processes involving the Tenax trap

Conditioning:

All conditioning is done according to the method proposed by Ward as it seems to work fine:

The trap tube is packed with Tenax while two chunks of glass wool, on both sides of the Tenax resin, hold it in place within the tube. He (99.995%) is passed through the tube at 150ml/min for 2 hours while the tube is being heated to 150°C. After two hours the

heating is stopped yet He is allowed to flow through the trap until it reached room temperature.

Hot blank:

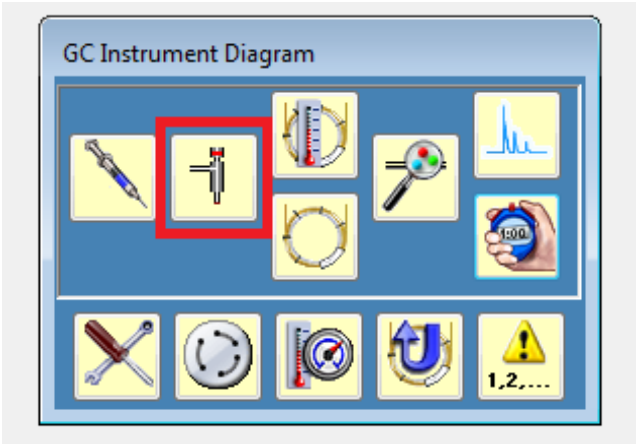
This type of analysis is done to make sure the Tenax trap is clean before and/or after a sample trap-release cycle.

1. Helium is passed through the system at 10 ml/min (±2ml/min). LN2 trap is installed at 6WV1 to catch any compound released from the Tenax.
2. The Tenax trap is heated to 150°C, make sure you record the time it took to reach 150°C.
3. The Tenax is held at 150°C for 3 minutes, make sure you record the maximum temperature the Tenax trap experienced.
4. After exactly 3 minutes the contents of the LN2 trap are injected to the the GC (Position "B" for 15 seconds).

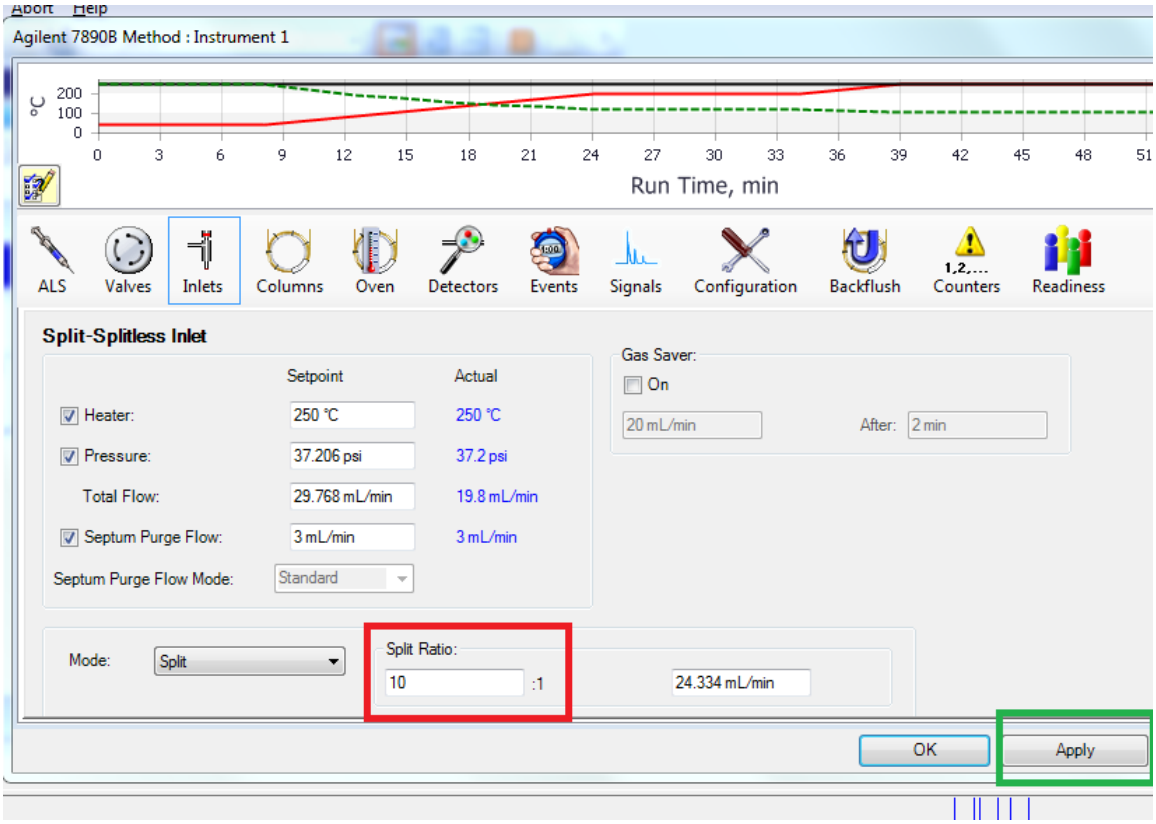
Successful analysis is one that has either no peaks at all or has peaks after ~30 minutes- it's a range not used in the natural gas project.

How to control split and valve timing:

Upon injecting via gas inlet or injection the split in the injector can be modified. From what me and Alex have tried this GC is capable of split in the range of 5-200. To change the split ratio choose the "GC Instrument Diagram" and press the injector icon:

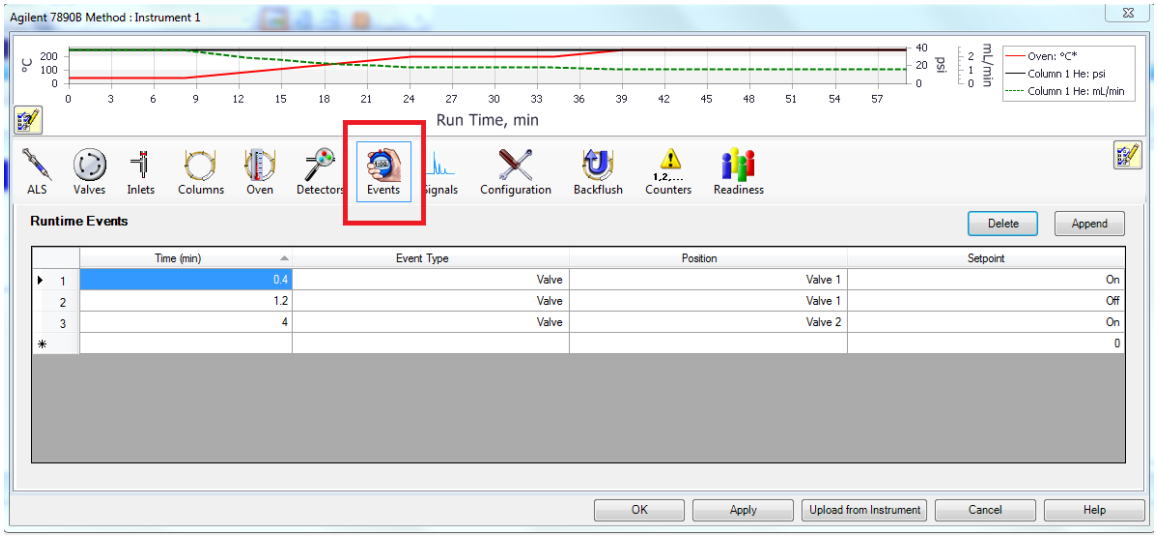


You will get a window in which the split ratio is presented and can be changed. After changing the split to the desired ratio make sure you press the “Apply” button to validate the new setting.



This GC is equipped with a dean switch which can be used to divert the column outlet between the available detector. All of our methods use TCD as the initial detector and the

column outlet is diverted to the FPD only at a later stage. The exact timing of the detector shift can be set via the “Events” section in the “GC Instrument Diagram” window:



In the “Events” section one can play with valve 1 and 2. Valve 1 is the sampling valve for gas in the inlet- at the standard setting it turns on at 0.4 minute (24 seconds) and then turns off at 1.2 minute (72 seconds)- when the valve is on the injected sample is being released from the internal loop to the column. Valve 2 is in charge of diverting the column outlet between the detectors. When valve 2 is off the outlet goes to the TCD detector, when its on the outlet goes to the FPD. The typical timing settings I use are:

Time of switching valve 2 to “on” position	Type of analysis
4	Running pure helium as blank with the intention of checking how the clean the GC is
6.5	Running pure methane or natural gas samples **with no H_2S**
8.5	Running gas samples that contain no more than few percent of H_2S (<~5%)

10	Running gas samples that contain high amounts of H_2S but also contain MeSH (elutes at ~10.5 minutes) Also good for blank methane experiments!